Trend-Following, Risk-Parity and the influence of Correlations[1]

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Abstract

Trend-following strategies take long positions in assets with positive past returns and short positions in assets with negative past returns. They are typically constructed using futures contracts across all asset classes, with weights that are inversely proportional to volatility, and have historically exhibited great diversification features especially during dramatic market downturns. However, following an impressive performance in 2008, the trend-following strategy has failed to generate strong returns in the post-crisis period, 2009-2013. This period has been characterised by a large degree of co-movement even across asset classes, with the investable universe being roughly split into the so-called Risk-On and Risk-Off subclasses. We examine whether the inverse-volatility weighting scheme, which effectively ignores pairwise correlations, can turn out to be suboptimal in an environment of increasing correlations. By extending the conventionally long-only risk-parity (equal risk contribution) allocation, we construct a long-short trend-following strategy that makes use of risk-parity principles. Not only do we significantly enhance the performance of the strategy, but we also show that this enhancement is mainly driven by the performance of the more sophisticated weighting scheme in extreme average correlation regimes.

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1. Introduction

Trend-following is a simple trading strategy that consists of long positions for upward trending assets and short positions for falling assets. This strategy profits when assets continue performing in-line with their most recent performance. In other words, this strategy aims to take advantage of return autocorrelation empirical patterns.[3]

Trend-following strategies are largely employed by systematic funds, like commodity trading advisor (CTA) and managed futures funds[4] (see Covel, 2009 for a broad overview), and are typically constructed using futures contracts across all asset classes[5] in an effort to increase diversification. The benefit from using futures contracts is two-fold: first, taking long and short positions using futures contracts is equally straightforward (in contrast, for instance, to using cash equity instruments) and second, the use of futures contracts allows the inclusion of non-equity contracts in the portfolio (e.g. trading commodities for investment purposes is typically done using futures).

The construction of a trend-following portfolio involves an important challenge, which is the choice of the weighting scheme that should be employed, given that contracts from different asset classes have very different risk-return profiles (a typical commodity or equity index contract is much more volatile than a government bond contract). An equal-weight allocation would result in a portfolio that would be dominated in terms of risk by the higher volatility assets, i.e. equities and commodities. Instead, the weighting scheme should make use of the relative riskiness of the contracts in order to allocate risk as evenly as possible across all constituents.

The typical choice is to employ inverse-volatility weights, so that all assets enter the portfolio with the same ex-ante volatility. For obvious reasons, this scheme is known as the volatility-parity scheme. This approach has been followed by the large majority of academic research papers focusing on the topic: e.g. Moskowitz, Ooi and Pedersen (2012), Hurst, Ooi and Pedersen (2012, 2013) and Baltas and Kosowski (2013, 2015). Importantly enough, as long as all pairwise correlations are equal, this weighting scheme splits the total portfolio volatility equally across all portfolio constituents.

Using a broad dataset of 35 futures contracts from all asset classes (energy, commodities, fixed income, foreign exchange and equities) we construct a volatility-parity trend-following strategy and document its superior performance relative to a

[3] Trend-following (also known as time-series momentum) is structurally different from the conventional cross-sectional winners-minus-losers momentum strategy of Jegadeesh and Titman (1993, 2001). The former is a strategy that takes a position in every asset of the investable universe, is not cash-neutral and, at the extreme, can be in a long-only or short-only state (if all assets have a positive or negative past return respectively); hence, it is a clear bet on the serial correlation of returns. Instead, the latter invests only in the extremes of the cross-section (e.g. top vs. bottom decile), it is –in theory– cash-neutral and its profitability can be either attributed to cross-sectional return dispersion premia or time-series return correlation (the recent paper by Asness, Moskowitz and Pedersen (2013) documents cross-sectional momentum patterns "everywhere"). For an analysis of the relationship between the two momentum strategies see Moskowitz, Ooi and Pedersen (2012) and Clare, Seaton, Smith and Thomas (2014a).

[4] Baltas and Kosowski (2013) show that futures-based trend-following strategies can explain large part of the return variation of CTA benchmark indices.

[5] Szakmary, Shen and Sharma (2010) study trend-following strategies in commodity markets, Burnside, Eichenbaum and Rebelo (2011) study carry and trend-following strategies in currency markets and finally in two recent papers Clare, Seaton, Smith and Thomas (2014a, 2014b) study both cross-sectional momentum and trend-following strategies in commodity markets and across broad market indices of different asset classes (equities, bonds, commodities and real estate) from a global asset allocation point of view.

long-only equivalent over a long history of more than 25 years (1988 to 2013). By employing long and short positions, the trend-following strategy benefits from (either upwards or downwards) trending markets and achieves in neutralising (at least unconditionally) the exposure to standard benchmark indices like the MSCI World Index or the S&P GSCI Index. The strategy benefits from the combination of different asset classes and delivers a Sharpe ratio of 1.31 compared to a 0.70 for the long-only equivalent over the entire sample period.

Contrary to the historical superior performance and following an impressive performance in 2008, the trend-following strategy has consistently delivered very poor performance in the post-crisis period (see also Hurst, Ooi and Pedersen 2012 and Baltas and Kosowski 2013). Between January 2009 and December 2013, a volatility-parity trend-following strategy delivers a Sharpe ratio of 0.31 against a Sharpe ratio of 0.59 for the long-only counterparty. What could have possibly gone wrong?

Following the introduction of the Commodity Futures Modernization Act (CFMA) in 2000, commodities have started becoming more correlated to each other as futures markets became accessible to investors as a way to hedge commodity price risk in what is often referred to as the *"financialisation of commodities"*.[6] More generally and more aggressively, following the recent financial crisis in 2008, assets from different asset classes (and not just commodities) have started exhibiting stronger co-movement patterns, with the diversification benefits being dramatically diminished.

In an environment of increased asset co-movement, the volatility-parity weighting scheme can be deemed a suboptimal choice. By ignoring the covariation between assets, volatility-parity fails to allocate equal amount of risk to each portfolio constituent. This is the reason why volatility-parity is also often called as naïve risk-parity (Bhansali, Davis, Rennison, Hsu and Li, 2012). Following these observations, one possible reason for the recent lacklustre performance of trend-following can be the suboptimal weighting scheme that ignores pairwise correlations (see e.g. Baltas and Kosowski, 2015). Our aim is to address this particular feature of the strategy and construct a portfolio that formally accounts for pairwise correlations.

At this stage, it is important to stress that the profitability of a trend-following strategy depends on two factors: (i) the existence of serial-correlation in the return series and (ii) the efficient combination of assets from various asset classes. It is obvious that the first factor is of utmost importance for the profitability of the strategy; non-existence of persistent price trends cannot be alleviated by a more robust weighting scheme. By amending the volatility-parity scheme in a way that accounts for pairwise correlations, we can only address any inefficiency in the risk allocation between portfolio constituents. However, it is reasonable to argue that a different portfolio allocation technique can only do so much.

In principle, an optimal allocation to risk that would also account for correlations would optimally over-weight assets, which correlate less with the rest of the universe and under-weight assets that correlate more with the rest of the universe in an effort to improve the overall portfolio diversification. This is the principle of the risk-parity

[6] The financialisation of commodities has recently been a very active research field. Indicatively, see the recent papers by Falkowski (2011), Irwin and Sanders (2011), Tang and Xiong (2012), Basak and Pavlova (2014), Boons, deRoon and Szymanowska (2014), Cheng and Xiong (2014) and Henderson, Pearson and Wang (2015) as well as references therein.

portfolio construction methodology (also known as the Equal Risk Contribution scheme). That is, to equate the contribution to risk from each portfolio constituent, after accounting for any pairwise correlation dynamics. Risk-parity has been a very popular portfolio construction technique during the last decades due to its remarkable performance (see for example[7] Anderson, Bianchi and Goldberg, 2012, and Asness, Frazzini and Pedersen, 2012) and has therefore been a topic of extensive research.[8]

Applying directly the risk-parity methodology to a trend-following strategy is not admissible, because risk-parity is defined as a long-only allocation framework.[9] Inspired by Jessop *et al.* (2013), we contribute to the literature by extending the conventional long-only risk-parity framework, in a way that also allows for short positions. To achieve this extension, we first generalise the conventional long-only risk-parity scheme into a risk-budgeting scheme, which, in turn, allows for the introduction of long and short positions in the overall portfolio.

The empirical question is whether this more sophisticated scheme can overcome the limitations of volatility-parity and consequently hedge against drawdowns experienced in high-pairwise-correlation states. Our findings show that the trend-following portfolio that employs risk-parity principles constitutes a genuine improvement to the traditional volatility-parity variant of the strategy. The Sharpe ratio of the strategy increases from 1.31 to 1.48 over the entire sample period (April 1988 – December 2013), but most importantly it more than doubles over the post-crisis period (January 2009 – December 2013) from 0.31 to 0.78. The improvement is both economically and statistically significant. A correlation event study shows that the improvement is mainly driven by the superior performance of the risk-parity variant of the strategy in extreme average correlation conditions.

The paper is organised as follows. Section 2 describes the construction of a typical trend-following portfolio using a volatility-parity weighting scheme. Section 3 presents the futures dataset that is used for the empirical analysis and Section 4 presents a thorough empirical evaluation of trend-following strategies, while highlighting their recent lacklustre performance. Section 5 provides the theoretical framework under which a long-only risk-parity framework can be extended into a long-short allocation and Section 6 reports the empirical benefits when this allocation is used for the trend-following strategy. Finally, Section 7 concludes.

[7] Both papers by Anderson *et al.* (2012) and Asness *et al.* (2012) employ inverse-volatility weights (volatility-parity), which they call "risk-parity" weights, for a stocks and bonds portfolio (2-asset portfolio). To avoid confusion, a risk-parity allocation for two assets degenerates mathematically into a volatility-parity allocation. Along these lines, their claim for "risk-parity" is valid as a special 2-asset case.

[8] The long list of papers includes Maillard, Roncalli and Teiletche (2010), Bhansali (2011), Inker (2011), Lee (2011), Chaves, Hsu, Li and Shakernia (2011, 2012), Bhansali, Davis, Rennison, Hsu and Li (2012), Leote de Carvalho, Lu and Moulin (2012), Lohre, Neugebauer and Zimmer (2012), Bernandi, Leippold and Lohre (2013), Lohre, Opfer and Orszag (2014), Fisher, Maymin and Maymin (2015) and Jurczenko, Michel and Teiletche (2015).

9 It is worth-highlighting that two recent papers by Clare, Seaton, Smith and Thomas (2014a, 2014b) claim to combine risk-parity with trend-following strategies, but in practice, they only employ conventional volatility-parity schemes that call "risk-parity". Similarly, Fisher *et al.* (2015) call "risk-parity" portfolio what effectively is a volatility-parity portfolio and reserve the term "equal risk contribution" for what we call "risk-parity".

2. Methodology

This section describes the steps for constructing a typical trend-following portfolio employs a volatility-parity weighting scheme.

2.1 Constructing a Trend-Following Strategy

Let N_t denote the number of available assets at time t. A trend-following (TF, henceforth) strategy involves taking a long or short position on each asset i, based on the sign of the past excess return over a prescribed lookback period that is typically equal to 12 months.[10] Let $w_t^{i,Gross}$ denote the gross (absolute) weight invested in asset i at time t. Trivially, $\sum_{i=1}^{N_t} w_t^{i,Gross} = 100\%$ and the return of the strategy is given by:

$$r_{t,t+1}^{TF} = \sum_{i=1}^{N_t} \underbrace{sign\left(r_{t-12,t}^{i}\right) \cdot w_t^{i,Gross}}_{w_t^{i,Net}} \cdot r_{t,t+1}^{i} \tag{1}$$

The net weights, denoted by $w_t^{i,Net}$, do not in practice add up to 100%, since they can take either positive or negative values.

Trend-following strategies are typically implemented using a constant-volatility[11] (CV, henceforth) overlay by targeting ex-ante a prescribed level of volatility σ_{TGT}. This requires employing dynamic leverage that is equal to the ratio between the running realised volatility of the unlevered trend-following strategy of equation (1), denoted by σ_t^{TF}, and the target level, σ_{TGT}. The generalised formulation of a constant-volatility trend-following ($CVTF$) strategy is therefore given by:

$$r_{t,t+1}^{CVTF} = \frac{\sigma_{TGT}}{\sigma_t^{TF}} \cdot \sum_{i=1}^{N_t} w_t^{i,Net} \cdot r_{t,t+1}^{i} \tag{2}$$

The dynamic leverage equals the ratio $\sigma_{TGT}/\sigma_t^{TF}$. As an example, if $\sigma_{TGT} = 10\%$ and at the end of some month the running volatility of the unlevered strategy is 5%, then all positions for the forthcoming month should be doubled (a 2x leverage ratio).

The final step is to define the functional form of the portfolio weights in equation (2).

2.2 Volatility-Parity Scheme

Given that trend-following strategies are formed across multiple asset classes, whose assets have very different risk profiles (as presented later in Figure 2), it is critical to determine a weighting scheme that assigns a weight to every asset that is a function of its underlying riskiness, in an effort to construct a strategy with a fairly balanced distribution of risk across assets and asset classes.

[10] In unreported results, we find that a 12-month horizon generates the largest Sharpe ratio for trend-following strategies across each asset class in line with Moskowitz *et al.* (2012) and Baltas and Kosowski (2013). See also Baltas, Jessop, Jones and Zhang (2013).

[11] A similar technique has been employed by Barroso and Santa-Clara (2014) and Daniel and Moskowitz (2014), who focus on cross-sectional winners-minus-losers momentum strategies. See also Hallerbach (2012, 2014).

The obvious and simplest choice is to employ inverse-volatility weights, so that all assets enter the portfolio with the same ex-ante volatility. For this reason, this weighting scheme is also called Volatility-Parity (VP, henceforth):

$$w_t^{i,Gross,VP} = \frac{1/\sigma_t^i}{\sum_{j=1}^{N_t} 1/\sigma_t^j}, \quad \forall i \tag{3}$$

This weighting scheme that has been used extensively in every academic study that focuses on trend-following strategies; see Moskowitz, Ooi and Pedersen (2012), Hurst, Ooi and Pedersen (2012, 2013) and Baltas and Kosowski (2013, 2015). It can be shown that VP can split the portfolio volatility equally across all portfolio constituents as long as all pairwise correlations are equal. In practice, as we document at a later section of the paper, the pairwise correlations between assets and asset classes are neither equal nor constant over time. Under such conditions, the distribution of risk of a VP scheme is not uniform and for that reason the scheme is also called naïve risk-parity (Bhansali *et al.*, 2012).

Going back to the construction of our benchmark trend-following strategy, substituting the VP weights back into equation (2) yields the return series of the Volatility-Parity Trend-Following ($VPTF$) strategy:

$$r_{t,t+1}^{VPTF} = \frac{\sigma_{TGT}}{\sigma_t^{TF}} \sum_{i=1}^{N_t} sign\left(r_{t-12,t}^i\right) \cdot \frac{\left(\sigma_t^i\right)^{-1}}{\sum_{j=1}^{N_t}\left(\sigma_t^j\right)^{-1}} \cdot r_{t,t+1}^i \tag{4}$$

For comparison purposes, we also construct a Long-Only (LO, henceforth) strategy benchmark as the inverse-volatility weighted portfolio of the assets ($VPLO$, henceforth), targeted again at the desired level of overall portfolio volatility:

$$r_{t,t+1}^{VPLO} = \frac{\sigma_{TGT}}{\sigma_t^{LO}} \sum_{i=1}^{N_t} (+1) \cdot \frac{\left(\sigma_t^i\right)^{-1}}{\sum_{i=1}^{N_t}\left(\sigma_t^i\right)^{-1}} \cdot r_{t,t+1}^i \tag{5}$$

3. Data Description

In order to construct trend-following strategies, we use Bloomberg daily closing futures prices for 35 contracts across all asset classes: 6 energy contracts, 10 commodity contracts, 6 government bond contracts, 6 FX contracts and 7 equity index contracts. The choice of the cross-section of contracts is presented in Table 1 and is considered to be fairly dispersed both across asset classes and global regions.

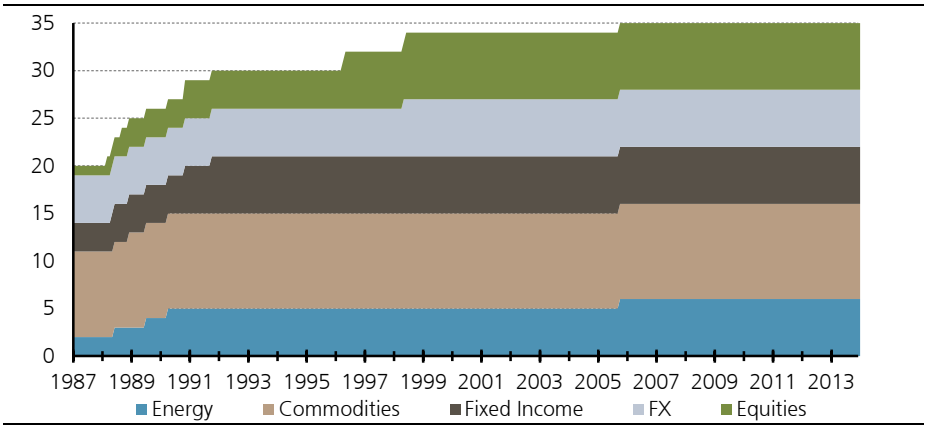
We restrict the sample period to start from January 1987, when all asset classes have at least one contract traded and the cross-section is relatively diverse with 20 contracts being traded in total. Figure 1 presents the evolution in the number of contracts for each asset class over time until the end of the sample period, December 2013.

Table 1: Dataset

ENERGY		COMMODITIES		FIXED INCOME		FX		EQUITIES	
Brent Crude	Jul-88	Cocoa	Jan-87	German Bobl 5Yr	Nov-91	AUD	Feb-87	Dax	Dec-90
Gas Oil	Aug-89	Copper	Jan-89	German Bund 10Yr	Dec-90	CAD	Jan-87	EuroStoxx 50	Jul-98
Gasoline	Nov-05	Corn	Jan-87	JGB 10Yr	Jan-87	CHF	Jan-87	FTSE 100	Mar-88
Heating Oil #2	Jan-87	Cotton #2	Jan-87	US T-Notes 5Yr	Jun-88	EUR	Jun-98	Kospi 200	Jun-96
Light Crude	Jan-87	Gold	Jan-87	US T-Notes 10Yr	Jan-87	GBP	Jan-87	Nasdaq	May-96
Natural Gas	May-90	Live Cattle	Jan-87	US T-Notes 30Yr	Jan-87	JPY	Jan-87	Nikkei	Oct-88
		Silver	Jan-87					S&P 500	Jan-87
		Soybeans	Jan-87						
		Sugar #11	Jan-87						
		Wheat	Jan-87						

Notes: The table reports the futures contracts that we use including the first month that each series is available. The dataset is retrieved from Bloomberg and the sample period ends in December 2013.

Figure 1: Number of Contracts per Asset Class



Notes: The figure presents the number of available futures contracts per asset class over time. The dataset is retrieved from Bloomberg and the sample period is January 1987 to December 2013.

It is important to note that futures contracts have, by their nature, two idiosyncratic features, which do not characterise spot cash equity instruments. First, futures contracts have finite life and are only traded for a short period of time before expiration. Second, futures contracts are zero-cost investments and, in theory, no capital is required to initiate a (long or short) position. In practice, entering into a new futures position implies posting collateral in form of an initial margin payment that is typically a small fraction of the prevailing futures price and a function of the contemporaneous riskiness (measured by volatility or VaR) of the underlying entity.

These specific features of futures contracts complicate the back-testing of futures-based trading strategies as continuous price-series have to be constructed and specific assumptions have to be put in place for the calculation of holding period returns as illustrated in Baltas *et al.* (2013) and Baltas and Kosowski (2015). We address these issues by using the generic continuous-price series provided by Bloomberg, which are constructed in such a way so that we always trade the most liquid contract (typically

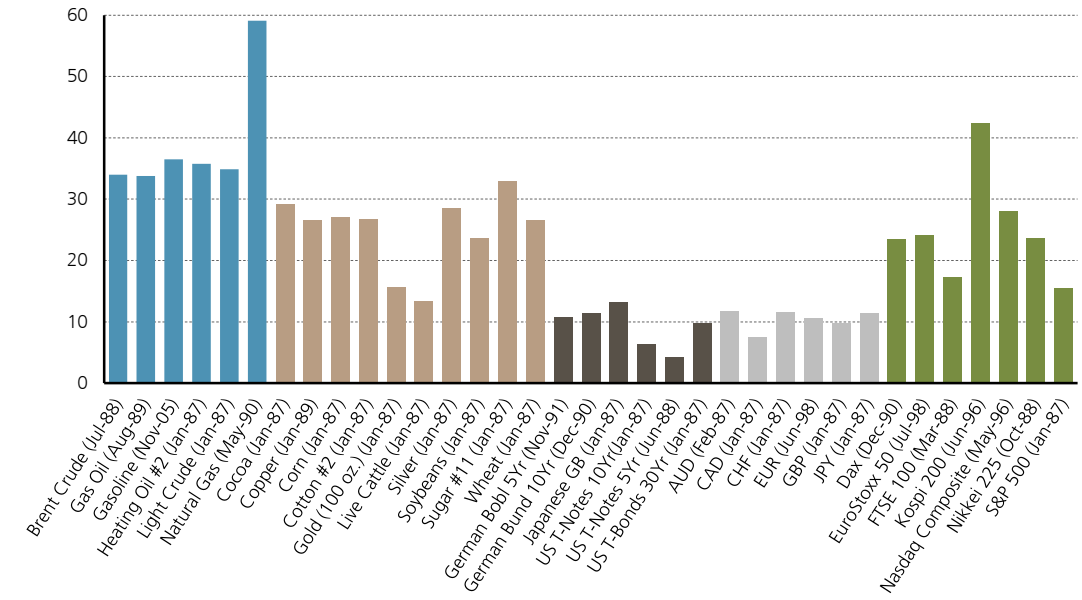
the "front" contract), and calculate for each futures contract fully-collateralised monthly returns in excess of the prevailing risk-free rate using the formula[12]:

$$r_{t,t+1} = \frac{F_{t+1} - F_t}{F_t} \tag{6}$$

where F_t and F_{t+1} denote the futures price at the end of months t and $t+1$.

Figure 2 presents annualised volatilities of all the assets in our dataset. What easily stands out is the large cross-sectional dispersion in volatilities, with fixed income contracts exhibiting traditionally the lowest volatilities. At the other end of the distribution, energy contracts are the most volatile contracts in the cross-section.

Figure 2: Asset Volatilities



Notes: The figure presents the unconditional volatility for each asset of the dataset. The colouring scheme separates the five asset classes (energy, commodities, fixed income, FX rates and equities). The legend states the starting month for each asset.

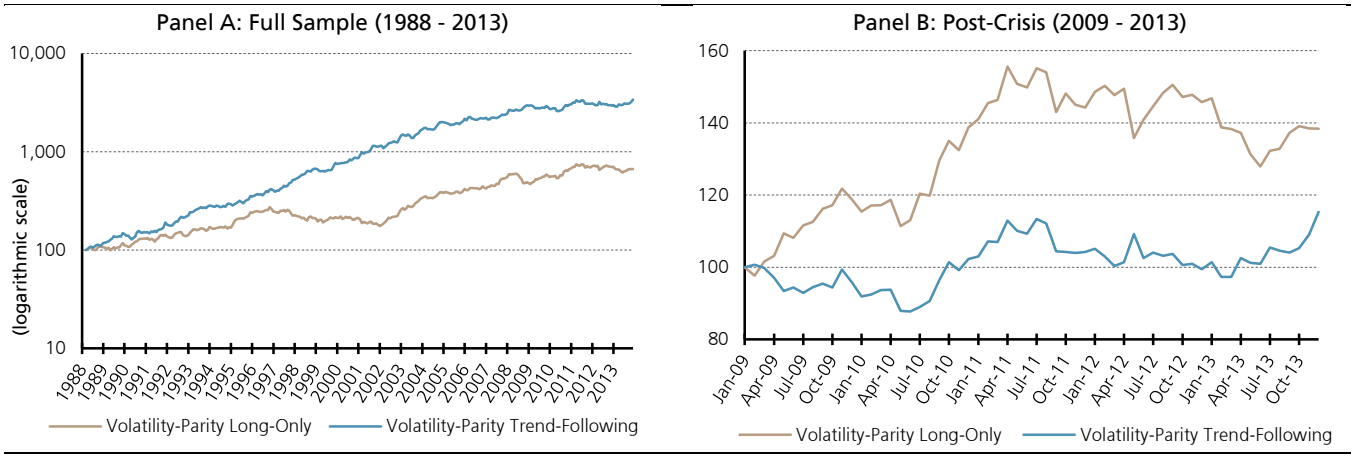
4. Performance Evaluation of Trend-Following

We start our empirical analysis by evaluating the performance of the trend-following strategy over the entire sample period and then focus on the most recent post-crisis period (2009-2013). The portfolio is rebalanced every month, when new trend-following signals are generated using the past 12-month performance of the assets. The weighting scheme (volatility-parity) is estimated using a window of the most recent 90 days until the end of each month. Finally, the strategy targets a 10% level of volatility; this requires an estimate of the running volatility of the unlevered strategy, which is estimated using the most recent 60 daily returns. The initial training period of the strategy is 12 months (for the signal generation) and another 60 business days for the calculation of the initial leverage ratio. Overall, this amounts to 15 months and therefore our sample period starts in April 1988.

[12] This approach in estimating returns of futures contracts is fairly standard in the academic literature. Indicatively, see de Roon, Nijman and Veld (2000), Moskowitz, Ooi and Pedersen (2012) and Baltas and Kosowski (2013, 2015).

Figure 3 presents the cumulative returns of a *VP* trend-following strategy and of its long-only variant across the two periods. Full-sample performance statistics as well as correlations with major benchmark indices of asset classes are reported in Table 2.

Figure 3: Cumulative Excess Returns of Volatility-Parity Long-Only and Trend-Following Strategies



Notes: The figure presents the cumulative returns of a long-only strategy and a trend-following strategy that employ a volatility-parity weighting scheme estimated using the past 90 days. The sample period is from April 1988 to December 2013 in Panel A and from January 2009 to December 2013 in Panel B.

Table 2: Volatility-Parity Long-Only and Trend-Following Strategies

Performance Statistics	*VPLO*	*VPTF*	Correlations	*VPLO*	*VPTF*
Ann. Geometric Mean (%)	7.63	14.67	*Commodity Benchmark:*		
t-statistic (Newey-West)	3.25	6.71	**- S&P GSCI Commodity Index**	0.58	0.11
Ann. Volatility (%)	11.54	10.96	*Fixed Income Benchmark:*		
Skewness	-0.12	0.38	**- JPM Global Bond Index**	0.69	0.31
Kurtosis	3.09	3.27	*FX Benchmarks:*		
Maximum Drawdown (%)	35.70	14.20	**- Trade-Weighted USD Index**	-0.54	-0.14
Sharpe Ratio (annualised)	0.70	1.31	**- USD/JPY**	-0.45	-0.18
Sortino Ratio (annualised)	1.13	2.81	*Equity Benchmarks:*		
Calmar Ratio	0.21	1.03	**- MSCI World (DM)**	0.52	-0.01
Monthly Turnover (%)	11.51	31.69	**- MSCI Emerging Markets**	0.37	-0.05

Notes: The table reports performance statistics and correlations with various benchmark indices (retrieved from Bloomberg) using monthly returns for the volatility-parity long-only strategy (*VPLO*) and for the volatility-parity trend-following strategy (*VPTF*) across all contracts. The t-statistic of the mean return is calculated using Newey and West (1987) standard errors. The Sortino ratio is defined as the annualised arithmetic mean return over the annualised downside volatility. The Calmar ratio is defined as the annualised geometric mean return over the maximum drawdown. The sample period is from April 1988 to December 2013.

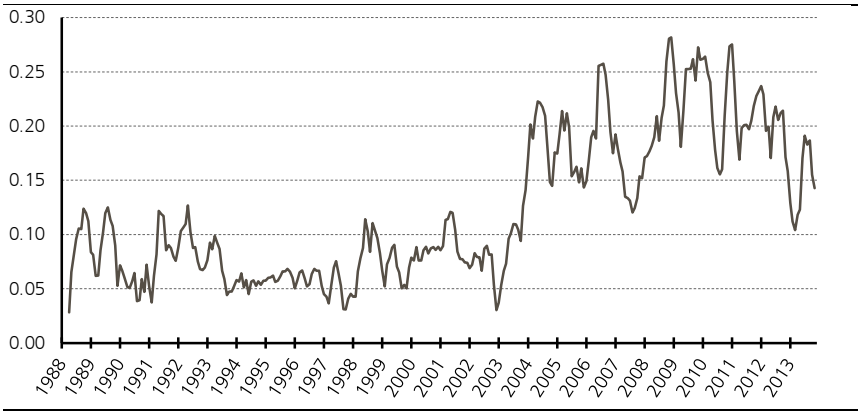
Over the entire sample period, the outperformance of the trend-following strategy is largely pronounced. The strategy exhibits a Sharpe ratio that is twice as big as that of the long-only strategy (1.31 vs. 0.70). In unreported results, we find that trend-following strategies within each asset class deliver Sharpe ratios between 0.58 and 0.71 over the same sample period, which means that the combination of different asset classes leads to a substantial improvement in the performance of the strategy. A detailed examination of trend-following patterns across various asset classes is presented in Moskowitz *et al.* (2012) and Baltas *et al.* (2013).

Most importantly, the trend-following strategy exhibits very low correlations with the various benchmark indices,[13] whereas the long-only strategy bears, by construction, strong directional bets and therefore large correlations with these indices. This piece of evidence justifies the use of trend-following strategies as diversification vehicles.

Contrary to the above evidence, when we shift our attention to the most recent post-crisis period, between January 2009 and December 2013, we find that the trend-following strategy has dramatically underperformed (see Figure 3, Panel B). The Sharpe ratio in the aftermath of the global financial crisis has been 0.31, compared to 0.59 of the long-only strategy. This recent lacklustre performance for a strategy that has historically delivered very strong returns across both up and down markets over several decades has been largely highlighted both in academic studies (e.g. Baltas and Kosowski, 2013) as well as in the press. What could have possibly gone wrong?

One of the most prevalent claims for this recent lacklustre performance of trend-following has been the post-crisis increased level of pairwise correlations (see e.g. Baltas and Kosowski, 2015). In an environment of increased correlations, diversification benefits diminish and assets are clustered into "Risk-On" and "Risk-Off" subsets. Figure 4 presents a 90-day rolling estimate of the average pairwise correlation across all futures contracts of our dataset. It is obvious that the level of pairwise correlations has significantly shifted during the last decade of the sample period, exhibiting one of the most prevalent increases in record during 2008.

Figure 4: Rolling Average Pairwise Correlation across all assets

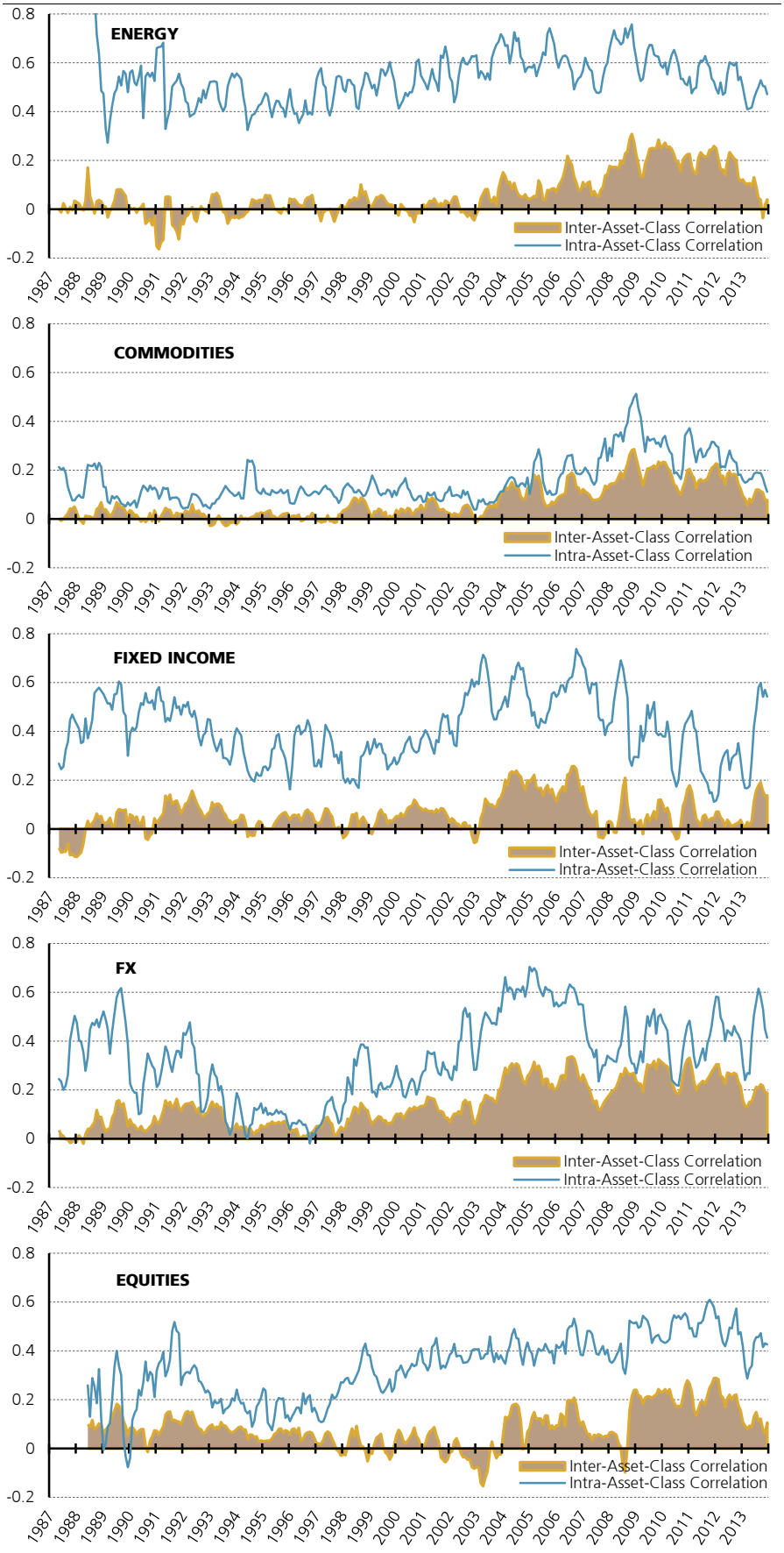


Notes: Monthly average pairwise correlation across all contracts using a 90-day rolling estimation window. The sample period is from April 1988 to December 2013.

In order to comprehend these dramatic shifts in asset pairwise correlations, Figure 5 presents the average correlation between assets of the same asset class (intra-asset class correlations) and the average correlation between assets of one asset class with the assets of all other asset classes (inter-asset class correlations). The evidence shows that the recent dramatic shift in average pairwise correlation is largely driven by the fact that all inter-asset class correlations have significantly increased during the most recent decade. The only exception is the correlation of fixed income assets with the rest of the asset classes. These plots illustrate clearly the risk-off (fixed income) and risk-on (remaining asset classes) regrouping of assets after the recent financial crisis.

[13] It is worth commenting that the trend-following strategy exhibits an almost zero unconditional correlation with the MSCI World Index (point estimate is -0.01). In unreported results, we find that this seemingly uncorrelated pair bears interesting non-linear (higher-order) correlation dynamics and in particular, the trend-following strategy exhibits strong positive returns in large positive or negative states of the market in line with Moskowitz *et al.* (2012).

Figure 5: Intra and Inter Asset-Class Average Pairwise Correlations



Notes: 90-day average pairwise correlation at the inter-class and intra asset-class level.

4.1 Volatility-Parity ignores Pairwise Correlations

Following the documentation of the correlation patterns, we return to the performance of the trend-following strategy. The weighting scheme that has been employed so far for the analysis is the volatility-parity scheme, which, as already highlighted, completely ignores the correlation of the assets. We therefore hypothesise that under an increased correlation environment, like the post-crisis era, volatility-parity is not an optimal risk allocation methodology.

Before evaluating this hypothesis, we should first illustrate how volatility-parity has allocated weights and risk across the five asset classes over time. The gross weight per asset class is simply calculated by summing up the individual gross weights of the constituents of each asset class at the end of each month, i.e. $\sum_j |w_t^j|$, where the summation is performed only across the assets within each asset class.

In order to calculate the percentage risk allocation per asset class, we should first define the so-called marginal contribution to risk (MCR, henceforth) for each asset. This is defined as the partial derivative of portfolio volatility at any point in time, σ_t^P, with respect to the contemporaneous weight of each asset, w_t^i, or in other words the change in portfolio volatility for a small (hence, marginal) change in the asset weight:

$$MCR_t^i = \frac{\partial \sigma_t^P}{\partial w_t^i} \tag{7}$$

It is easy to show that the *MCR*'s satisfy the following identity[14]:

$$\sum_{j=1}^{N_t} w_t^j \cdot MCR_t^j = \sigma_t^P \tag{8}$$

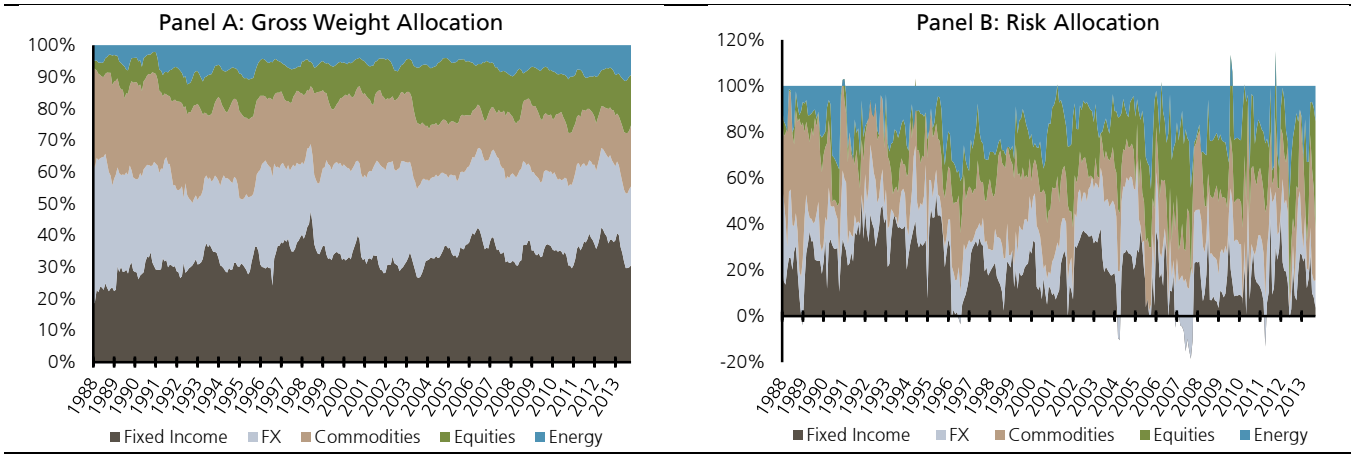
Given this definition, the percentage contribution to risk from each asset class at the end of each month is calculated by summing the weighted MCR of each asset in the asset class, normalised by the total portfolio volatility, i.e. $\sum_j w_t^j \cdot MCR_t^j / \sigma_t^P$, where the summation is performed only across the assets within the same asset class. It is critical to notice that if all pairwise correlations are equal then the percentage contribution of each asset is simply $1/N_t$ and therefore the percentage contribution of each asset class would be proportional to the asset class size. Given that we have fairly balanced asset classes (6 energy contracts, 10 commodity contracts, 6 government bond contracts, 6 FX contracts and 7 equity index contracts) this would result in almost equal asset class contributions to portfolio risk.

Figure 6 presents the time evolution of the weight and risk allocation across the five asset classes. The former is fairly stable over time, due to the large persistence of volatility measures; the strategy allocates on average 33% in fixed income, 27% in FX, 21% in commodities, 12% in equities and 7% in energy). However, when translated into risk terms, this allocation is **nowhere close** an equal distribution across constituents. The risk allocation per asset class is largely unstable over time and more

[14] Contrast this with the fact that the weighted sum of volatilities typically exceeds the overall portfolio volatility due to the diversification benefit: $\sum_{j=1}^{N_t} w_t^j \cdot \sigma_t^j \geq \sigma_t^P$.

importantly, there are times during which some asset classes (with the given VP weights) have negative risk contribution (i.e. diversify risk away), like for fixed-income during 2007. At such times, it would be reasonable to increase the allocation to these asset classes, as this would lower portfolio risk.

Figure 6: Gross Weight and Risk Allocation for the Volatility-Parity Trend-Following Strategy



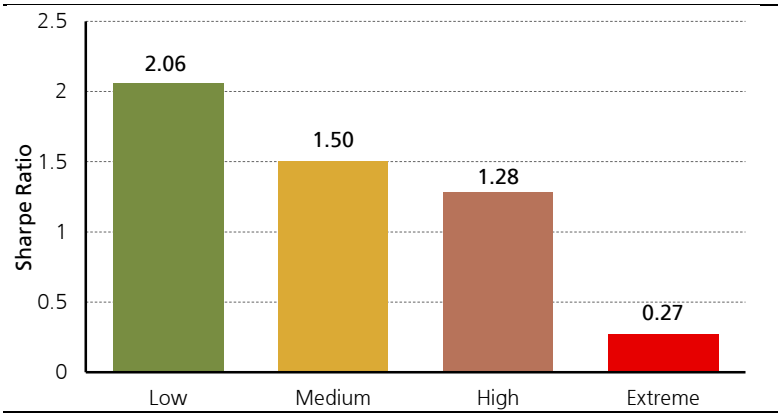
Notes: The figure presents the sum of gross weights (Panel A) and the percentage constitution to risk (Panel B) for each asset class over time when a volatility-parity weighting scheme is employed for a trend-following strategy. The sample period is from April 1988 to December 2013.

Evidently, the VP allocation does not equate the risk contribution from each asset class and this is solely due to the time-varying nature of the correlations between assets and asset classes as documented in Figure 5.

Going back to the hypothesis that the recent poor performance of trend-following could be potentially related to the suboptimal allocation of risk from the VP scheme, we next conduct a correlation event study that is presented in Figure 7. In particular, we first calculate the average pairwise correlation of the universe of all assets for each calendar month in our sample, using only the daily returns of the assets within each particular month. Next, we group all months of the dataset in four correlation regimes; low: less than 5%, medium: 5% to 10%, high: 10% to 20% and extreme: more than 20%. For each correlation regime, we estimate the Sharpe ratio of the trend-following strategy. The evidence is overwhelming. The performance of the VP trend-following strategy drops dramatically when the level of average pairwise correlation deviates significantly away from zero and into the positive territory and the drop is most dramatic as we move from a high correlation regime (Sharpe ratio of 1.28) to an extreme one (Sharpe ratio of 0.27).

This performance drop can be attributed to two possible reasons: (i) the absence of strong price trends in high correlation regimes and/or (ii) the sub-optimal distribution of risk to portfolio constituents by the volatility-parity weighting scheme. The objective of the paper is to focus solely on the portfolio construction implications for trend-following strategies and therefore address the extent to which a more sophisticated portfolio construction methodology that accounts for the time-varying nature of correlations, can improve the diversification of the portfolio in higher correlation regimes and therefore improve its risk-adjusted performance.

Figure 7: Correlation Event Study



Notes: The figure presents the annualised Sharpe ratio of a volatility-parity trend-following strategy for four different regimes of average pairwise correlation: low (less than 5%; 49 months), medium (5% to 10%; 109 months), high (10% to 20%; 92 months) and extreme (above 20%; 59 months). The sample period is from April 1988 to December 2013.

5. Risk-Parity Principles

This section outlines the steps to extend the risk-parity principles to a long-short allocation. This involves an intermediate step of defining a long-only and subsequently a long-short risk-budgeting framework.

5.1 Risk-Parity

Risk-Parity (RP, henceforth) constitutes the extension to the volatility-parity weighting scheme and its objective is to distribute the total portfolio risk (volatility) equally across the portfolio constituents, after accounting for pairwise correlations. Using the property of equation (8), this objective is equivalent to equating the weighted marginal contribution to risk of each constituent:

$$w_t^{i,RP} \cdot MCR_t^i = constant, \quad \forall i \tag{9}$$

where MCR is defined as in equation (7). The constant in the above equation can be trivially shown to be equal to the $\frac{1}{N_t}$th of the portfolio volatility.

The solution to the RP objective does not come in closed-form (unless all pairwise correlations are equal, in which case the solution boils down to VP), but instead through an optimisation. Following Jessop *et al.* (2013), this can be attained by maximising the sum of logarithmic weights, subject to a risk constraint of target volatility σ_{TGT} for the whole portfolio:[15]

Long-Only Risk-Parity:

Maximise:

$$\sum_{i=1}^{N_t} log(w_t^i)$$

Subject to: $\sqrt{\boldsymbol{w}_t' \cdot \boldsymbol{\Sigma}_t \cdot \boldsymbol{w}_t} \leq \sigma_{TGT}$ (10)

[15] Logarithmic weights for the formulation of risk-parity portfolios are also used by Kaya (2012), Kaya and Lee (2012) and Roncalli (2014).

where $\boldsymbol{w}$ denotes the vector of weigths and $\boldsymbol{\Sigma}$ denotes the variance-covariance matrix of the universe, both evaluated at time t. It's easy to show (see the Appendix) that the Lagrangian of the optimisation results in the risk-parity objective of equation (9). This optimisation is solved in practice, using a non-linear optimiser with the initial guess for the weight vector being the VP solution, because this is exactly the point of convergence of RP weights when all correlations are equal:

$$w_t^{i,RP,initial} = w_t^{i,VP} = \frac{1/\sigma_t^i}{\sum_{j=1}^{N_t} 1/\sigma_t^j}, \quad \forall i \tag{11}$$

The final point that should be made is that the risk-parity portfolio weights that come out of the optimisation do not typically add up to 100%. Rescaling the resulting weights post-optimisation is admissible, because volatility, as a measure of risk, exhibits positive homogeneity (see the Appendix).

Risk-parity constitutes one of the most popular risk-based portfolio construction methodologies, however, it has only been defined for a long-only allocation; the objective function in the optimisation does not allow for negative weights (due to the logarithm) and, in fact, the optimisation sets a natural bound at zero for all weights. The risk-parity portfolio will always have strictly positive weights for all the assets.

For our purpose, which is the application of risk-parity principles to a trend-following strategy, the above formulation cannot be used. Using VP weights, it's straightforward to calculate gross weights that are inversely proportional to the asset volatilities and then invert the weights for these assets that we require a short position. However, solving a long-only risk-parity framework and subsequently inverting these weights is not admissible. The correlation structure of a long-only universe is very different to the correlation structure on a long-short universe. A short position on a particular asset means that all correlations of this asset with the rest of the universe switch sign. It is therefore signed correlations that should be used for the determination of a risk-parity long-short portfolio. Simply inverting the long-only solution for the assets with a short position is completely incorrect. We need a proper long-short framework. In order to introduce this, we first present the concept of long-only risk-budgeting.

5.2 Risk-Budgeting

The long-only risk-parity paradigm can be generalised to a long-only risk-budgeting (RB) framework, under which each portfolio constituent contributes an amount to the overall portfolio volatility that is proportional to a certain positive asset-specific score, denoted by s_t^i; these scores can be for instance the ranks from a customised screen of the universe. As an example, if $s_t^i = 2s_t^j$ for two assets i and j, then the objective is for asset i to have a weighted MCR that is twice as big as that of asset j. Hence, the RB objective is:

$$w_t^{i,RB} \cdot MCR_t^i \propto s_t^i, \quad \forall i \tag{12}$$

This objective can be attained by solving a modified version of the optimisation (10) as also shown in Kaya and Lee (2012):

Long-Only Risk-Budgeting:

Maximise:

$$\sum_{i=1}^{N_t} s_t^i \cdot log\left(w_t^i\right)$$

Subject to: $\sqrt{\boldsymbol{w}_t' \cdot \boldsymbol{\Sigma}_t \cdot \boldsymbol{w}_t} \leq \sigma_{TGT}$ (13)

Indeed, the Lagrangian of this optimisation coincides with equation (12). The key difference to the original RP framework is the introduction of the asset-specific score in the objective function. Solving this optimisation is again fairly straightforward, using an initial guess that is a score-adjusted variant of the VP weights:

$$w_t^{i,RB,initial} = \frac{s_t^i / \sigma_t^i}{\sum_{j=1}^{N_t} s_t^j / \sigma_t^j}, \quad \forall i \tag{14}$$

The long-only risk-budgeting framework is the intermediate step that we need in order to introduce the long-short risk-parity portfolio construction methodology.

5.3 Long-Short Risk-Budgeting

The interesting question is whether we can allow the risk-specific scores to be negative. Motivated by the arguments of Jessop *et al.* (2013), if the scores are estimates or views of expected returns of the assets, then these should be allowed to be negative. In other words, a negative view for a certain asset can be used as a signal for a short position. Based on this, we extend the risk-budgeting framework to a long-short risk-budgeting framework by allowing the asset-specific scores to take negative values and therefore instruct short positions. Along these lines:

$$sign\left(w_t^{i,Net,RB}\right) = sign\left(s_t^i\right), \quad \forall i \tag{15}$$

Given that the calculation of MCR encompasses the sign of the weight (a long position with a positive MCR implies that a negative position of the same size will have a negative MCR), the objective of the long-short risk-budgeting becomes:

$$w_t^{i,Net,RB} \cdot MCR_t^i \propto \left|s_t^i\right|, \quad \forall i \tag{16}$$

The only obvious difference to equation (12) is the introduction of an absolute value to the scores. However, there exist more fundamental, yet subtle, differences. Both the asset weight and the MCR are now signed, i.e. contain information about the type of position that is prescribed for the asset by the sign of the score.

In order to solve for this objective, we reformulate the optimisation as follows:

Long-Short Risk-Budgeting:

Maximise:

$$\sum_{i=1}^{N_t} \left|s_t^i\right| \cdot log\left(\left|w_t^i\right|\right)$$

Subject to: $\sqrt{\boldsymbol{w}_t' \cdot \boldsymbol{\Sigma}_t \cdot \boldsymbol{w}_t} \leq \sigma_{TGT}$ (17)

As before, the Lagrangian of this optimisation coincides with equation (16). In this formulation, both the score and the weight that feed into the objective function bear an absolute value. The absolute value of the weight is necessary so that the logarithm is also defined for short positions. The most important point is that the asset-specific score and the respective weight agree always in their signs. Along these lines, the objective function pushes the positive weights away from zero towards the positive territory, with the relative effects being more aggressive for assets with larger (positive) scores and equivalently pushes the negative weights away from zero towards the negative territory, with the effects being more aggressive for assets with larger (in absolute value, yet negative) scores.

In order for the above methodology to operate as explained, the initial weights for each position must match the sign of the respective scores. As long as this is the case, then the signs of the weights will not flip during the optimisation; instead, the signs of the weights are preserved due to the mathematical formulation of the optimisation and the use of the logarithm. The role of the optimisation is only to scale the weights, following risk-budgeting principles, while preserving their signs. The initial net weights can be deduced from the long-only RB approach after incorporating an absolute value in the denominator, so that the gross weights sum up to 100%:

$$w_t^{i,Net,RB,initial} = \frac{s_t^i / \sigma_t^i}{\sum_{j=1}^{N_t} |s_t^j| / \sigma_t^j}, \quad \forall i \tag{18}$$

As required, these initial weights will be positive for assets with positive scores and negative for assets with negative scores.

5.4 Trend-Following meets Risk-Parity

The long-short risk-budgeting framework is exactly what we need to introduce risk-parity principles to a trend-following strategy. Given that the sign of the asset-specific score instructs the type of position (long or short), we can set it equal to the trend-following signal, which is the sign of the past 12-month return of the asset at the end of each month:

$$s_t^i = sign\left(r_{t-12,t}^i\right), \quad \forall i \tag{19}$$

This choice achieves two goals at the same time. Not only does it formally incorporate the trend-following signals in the portfolio optimisation, but it also achieves the transition from the risk-budgeting framework back to risk-parity (i.e. equal risk contributions). This is because $\left|s_t^i\right| = \left|sign\left(r_{t-12,t}^i\right)\right| = 1$ and therefore the long-short risk-budgeting objective of equation (16) boils down to the conventional risk-parity objective:

$$w_t^{i,Net,RP} \cdot MCR_t^i \propto 1 \Rightarrow w_t^{i,Net,RP} \cdot MCR_t^i = constant, \quad \forall i \tag{20}$$

The difference to the original formulation is that now the optimisation allows for both long and short positions. In particular, the optimisation problem (17) simplifies into:

Long-Short Risk-Parity:

Maximise:

$$\sum_{i=1}^{N_t} log\left(|w_t^i|\right)$$

Subject to: $\sqrt{\boldsymbol{w}_t' \cdot \boldsymbol{\Sigma}_t \cdot \boldsymbol{w}_t} \leq \sigma_{TGT}$ (21)

and the initial guess for the solution, which follows naturally from equation (18) coincides with the net long-short VP scheme that we have used so far for trend-following strategy of equation (4):

$$w_t^{i,Net,RP,initial} = w_t^{i,Net,VP} = sign\left(r_{t-12,t}^i\right) \cdot \frac{1/\sigma_t^i}{\sum_{j=1}^{N_t} 1/\sigma_t^j}, \quad \forall i \tag{22}$$

Using the net weights that come out of the risk-parity trend-following optimiser, we finally get the Risk-Parity Trend-Following ($RPTF$) strategy:

$$r_{t,t+1}^{RPTF} = \frac{\sigma_{TGT}}{\sigma_t^{TF}} \cdot \sum_{i=1}^{N_t} w_t^{i,Net,RP} \cdot r_{t,t+1}^i \tag{23}$$

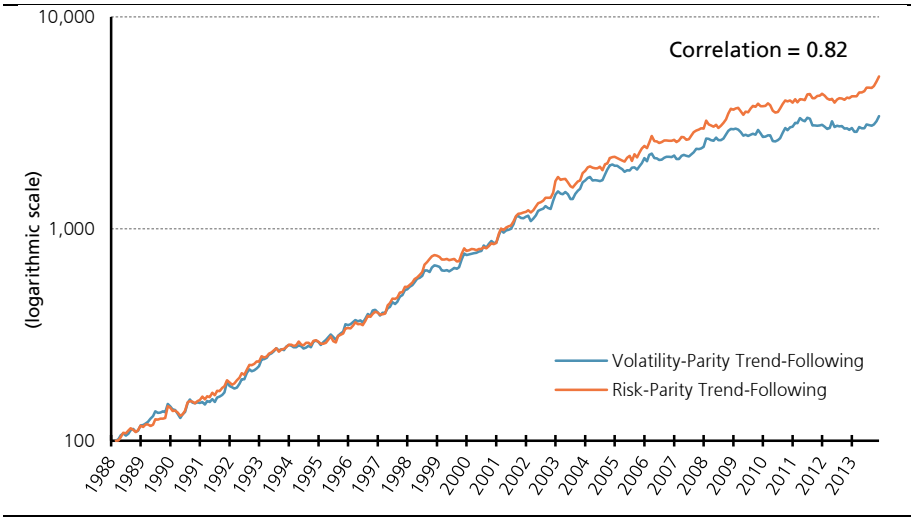
6. Performance Evaluation of Risk-Parity Trend-Following

Figure 8 presents the cumulative returns of a trend-following strategy that uses either a VP scheme or the just introduced RP long-short scheme. Both weighting schemes are dynamically estimated using a window of 90 business days up until the end of each month and both strategies employ a 10% target level of volatility. Full-sample performance statistics as well as correlations with major benchmark indices are reported in Table 3.

Visually, these two variants of trend-following exhibit a large degree of co-movement, and a full-sample correlation of 0.82, given that they employ the same set of long and short positions at the end of each month. They only differ in the weighting scheme. Importantly enough, up until 2003, the two lines are one and the same. This period has been characterised by relatively low and insignificant correlations between the asset classes (as shown in Figure 4), and therefore volatility-parity and risk-parity solutions are expected to be statistically and numerically indistinguishable.

However, post-2004, when correlations between different asset classes increase almost uniformly, the RP allocation system under-weights in relative terms the assets that are more correlated on average with the universe and accordingly over-weights the assets that have low average correlation with the universe and therefore exhibit larger diversification properties. This differentiation in the weight allocation appears to be driving the outperformance of the risk-parity solution over the most recent decade and most importantly during the post-crisis period.

Figure 8: Cumulative Excess Returns of Trend-Following Strategies



Notes: The figure presents the cumulative returns of a volatility-parity trend-following strategy and a risk-parity trend-following strategy. The weighting schemes are estimated using the past 90 days. Both strategies target a volatility of 10%. The scale is logarithmic. The sample period is between April 1988 and December 2013.

Table 3: The Variants of Trend-Following Strategies

Performance Statistics	***VPTF***	***RPTF***	**Correlations**	***VPTF***	***RPTF***
Annualised Geometric Mean (%)	14.67	16.61	*Commodity Benchmark:*		
t-statistic (Newey-West)	6.71	8.05	**- S&P GSCI Commodity Index**	0.11	0.07
Annualised Volatility (%)	10.96	10.86	*Fixed Income Benchmark:*		
Skewness	0.38	0.57	**- JPM Global Bond Index**	0.31	0.21
Kurtosis	3.27	3.83	*FX Benchmarks:*		
Maximum Drawdown (%)	14.20	10.67	**- Trade-Weighted USD Index**	-0.14	-0.09
Sharpe Ratio (annualised)	1.31	1.48	**- USD/JPY**	-0.18	-0.13
Sortino Ratio (annualised)	2.81	3.47	*Equity Benchmarks:*		
Calmar Ratio	1.03	1.56	**- MSCI World (DM)**	-0.01	0.00
Monthly Turnover (%)	31.69	57.74	**- MSCI Emerging Markets**	-0.05	0.00

Notes: The table reports various performance statistics and correlations with various benchmark indices (retrieved from Bloomberg) using monthly returns for the volatility-parity trend-following strategy ($VPTF$) and for the risk-parity trend-following strategy ($RPTF$). The t-statistic of the mean return is calculated using Newey and West (1987) standard errors. The Sortino ratio is defined as the annualised arithmetic mean return over the annualised downside volatility. The Calmar ratio is defined as the annualised geometric mean return over the maximum drawdown. The sample period is from April 1988 to December 2013.

Over the entire sample period, the *RP* allocation improves the performance of the trend-following strategy, with the Sharpe ratio increasing from 1.31 to 1.48. The performance improvement becomes more pronounced for performance ratios that account for downside risk, like the Sortino ratio (arithmetic mean return over annualised downside volatility) and the Calmar ratio (geometric mean return over maximum drawdown). Interestingly, the already low correlations with the various benchmark indices fall even further in absolute value.

In order to test whether this improvement in the performance is genuine and statistically strong, we calculate two-sided and one-sided p-values for a paired signed-rank Wilcoxon (1945) test. The equality in the average return between the two variants of the trend-following strategy is rejected with a two-sided p-value of 6.92% and more strongly with a one-sided p-value (in favour of the risk-parity variant) of

3.46%. In other words, the different weighting scheme results in a genuine and statistically strong improvement for the trend-following strategy.

The outperformance of the RP trend-following strategy against its VP variant becomes significantly more pronounced when we focus on the post-crisis period, 2009-2013. Table 4 reports performance statistics for the two strategies over this five-year period. In short, RP revives trend-following, achieving a statistically significant average return (t-statistic of 1.73 compared to the insignificant t-statistic of 0.78 for the VP variant) and Sharpe ratio of 0.78 compared to just 0.31 for the VP variant.[16] Focusing on the downside, the benefit is even more pronounced with the Calmar ratio (ratio between the geometric mean return and the maximum drawdown) almost quadrupling from 0.20 to 0.77.

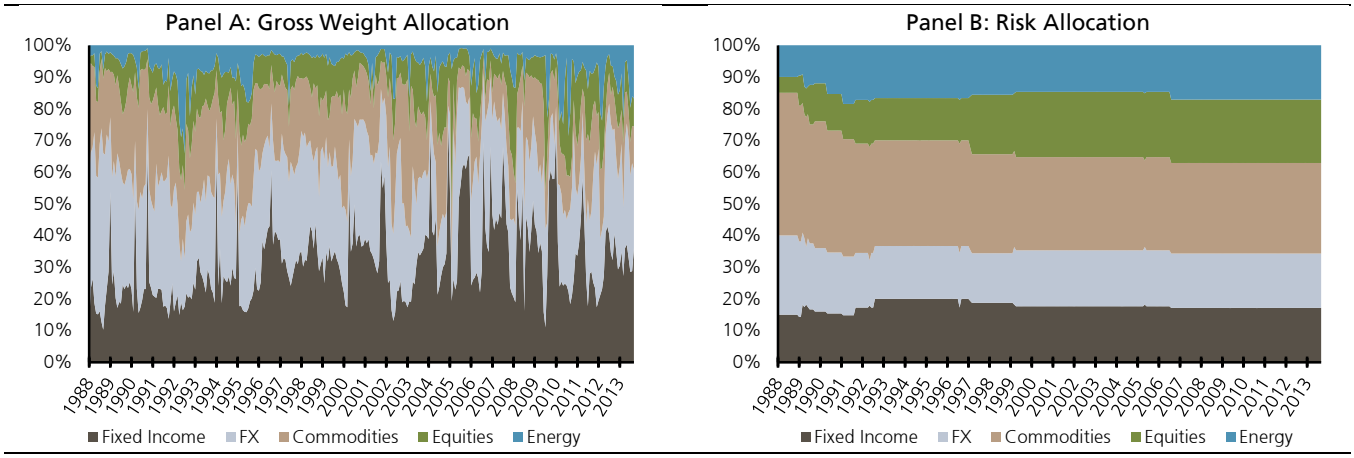
Table 4: The Variants of Trend-Following Strategies over 2009-2013

	$VPTF$	$RPTF$
Annualised Geometric Mean (%)	2.82	7.27
t-statistic (Newey-West)	0.78	1.73
Annualised Volatility (%)	10.77	9.60
Skewness	0.12	-0.06
Kurtosis	3.01	2.48
Maximum Drawdown (%)	14.20	9.49
Sharpe Ratio (annualised)	0.31	0.78
Sortino Ratio (annualised)	0.49	1.35
Calmar Ratio	0.20	0.77

Notes: The table reports various performance statistics for the volatility-parity trend-following strategy ($VPTF$) and the risk-parity trend-following strategy ($RPTF$) over the period between January 2009 and December 2013.

In order to draw parallels against volatility-parity, Figure 9 presents the absolute weight and risk allocation across the five asset classes for the risk-parity strategy. This figure should be studied alongside Figure 6.

Figure 9: Gross Weight and Risk Allocation for the Risk-Parity Trend-Following Strategy



Notes: The figure presents the sum of gross weights (Panel A) and the percentage constitution to risk (Panel B) for each asset class over time when a risk-parity weighting scheme is employed for a trend-following strategy. The sample period is from April 1988 to December 2013.

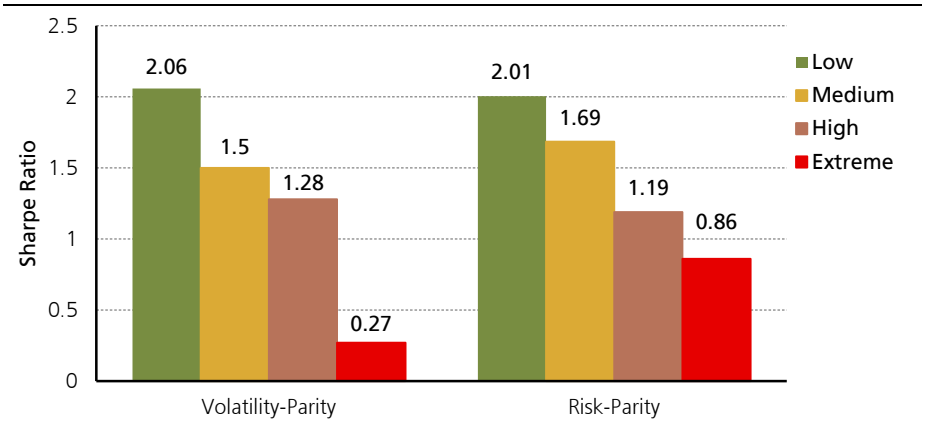
[16] The two-sided and one-sided p-values of a paired Wilcoxon (1945) signed-rank test on the equality between the mean return of the RP trend-following strategy and the VP trend-following strategy during the 2009-2013 period is 5.80% and 2.90% respectively, showing that the return of the RP strategy is statistically larger.

Evidently, by taking into account the pairwise correlations, risk-parity succeeds in ex-ante equating the risk contribution of each asset. The only reason that Panel B of Figure 9 exhibits (any) risk allocation shifts between asset classes is due to the fact that the overall size of the number of available contracts N_t is not constant over time.

In order to maintain the equal-risk-contribution objective, the RP scheme shifts radically between asset classes, as shown in Panel A of Figure 9, due to the fast-changing correlation environment. As an example, the gross allocation in fixed income contracts ranges between 10% and 76% over time, contrary to the more stable allocation under VP that ranges between 18% and 48% (from Panel A of Figure 6); the average allocation over time is very similar in both schemes (32% and 33% for RP and VP respectively). This clearly results in larger turnover for the RP variant of trend-following. Table 3 reports a full-sample monthly two-way turnover of 58% compared to 32% for the VP scheme. The additional turnover (and therefore additional transaction costs) should not subsume the genuine improvement in the risk allocation of the trend-following strategy, even though it might indeed reduce the after-costs benefit of the strategy.

Finally, in order to illustrate that the more sophisticated portfolio construction methodology can lead to superior performance especially in higher-correlation environments, we augment the correlation event study of Figure 7 with the performance of the risk-parity trend-following strategy under the four different correlation regimes. Figure 10 presents the results. It is evident that if there is any significant improvement in the performance of the strategy, this is mainly pronounced in extreme correlation environments, with the Sharpe ratio increasing from 0.27 up to 0.86. In such states of the market, the volatility-parity scheme is rendered suboptimal in its risk allocation across assets, whereas risk-parity, by taking into account the pairwise correlations, succeeds in delivering a more diversified allocation and therefore in delivering superior performance.

Figure 10: Correlation Event Study



Notes: The figure presents the annualised Sharpe ratio of a volatility-parity and a risk-parity trend-following strategy for four different regimes of average pairwise correlation: low (less than 5%; 49 months), medium (5% to 10%; 109 months), high (10% to 20%; 92 months) and extreme (above 20%; 59 months). The sample period is from April 1988 to December 2013.

A recent report by Baltas, Jessop, Jones and Zhang (2014) highlights that the added value of risk-parity in higher correlation environments is, in fact, due to larger dispersion of pairwise correlations between asset classes.

As a final point, it should be highlighted that even with the use of a risk-parity allocation, Figure 10 documents that trend-following suffers in higher-correlation environments. Evidently, this must be related directly to the non-existence of strong price trends in high correlation environments. As it has been shown, a more sophisticated weighting scheme can only do so much as far as portfolio diversification is concerned. Any further improvement of the strategy should look into the reasons why price trends might be less persistent in high correlation environments. Future research should address these claims.

From a different perspective, this performance drop can be rationalised using the Grinold and Kahn (2000) fundamental law of active management under which the information ratio equals the product of the manager's skill (information coefficient) and the square root of the breadth of the investable universe. Assuming a constant skill, when asset correlations increase, the breadth of the investable universe falls and this should expectedly cause a fall in the attainable information ratio.

7. Conclusion

Trend-following strategies have been very profitable historically and have constituted great diversification vehicles against market downturns like recently during the financial crisis of 2008. Their main source of profitability is related to the diversification benefit that stems from combining futures contracts across different assets classes, which have exhibited historically very low, if not insignificant, cross-correlations. A simple volatility-parity (inverse-volatility) weighting scheme has been historically considered appropriate in order to combine assets from different asset classes with very diverse risk profiles. Such a weighting scheme would distribute risk equally among constituents, should their correlations were constant over time.

Following an impressive performance in 2008, trend-following strategies have performed poorly over the most recent period (2009-2013), which has been characterised by dramatic increases in the correlations between different asset classes. We hypothesise that the volatility-parity weighting scheme leads to uneven and therefore suboptimal risk allocation under such conditions and this could be one of the reasons for the recent underperformance of trend-following.

A risk-parity weighting scheme can succeed in equating the contribution to the total portfolio risk from all portfolio constituents, after also accounting for correlations. However, its conventional formulation can only be applied to a long-only portfolio. Inspired by Jessop *et al.* (2013), we contribute to the literature by extending the conventional long-only risk-parity framework, in a way that also allows for short positions. This extension is necessary, if such a weighting scheme were to be employed in a trend-following portfolio.

The risk-parity trend-following portfolio constitutes a genuine improvement to the traditional volatility-parity variant of the strategy. The Sharpe ratio of the strategy increases from 1.31 to 1.48 over the entire sample period (April 1988 – December 2013), and more than doubles over the post-crisis period (January 2009 – December 2013) from 0.31 to 0.78. A correlation event study shows that the improvement is mainly driven from the superior performance of the strategy in extreme average correlation environments.

APPENDIX - Solving for Risk Parity

The risk parity methodology solves for portfolio weights at the end of each month, so that every asset is contributing the same amount of risk to the overall portfolio, i.e.

$$w_t^{RP,i} \cdot MCR_t^i = constant, \ \ \forall i \tag{24}$$

One can solve for this objective using the following non-linear constrained optimisation problem:

- Maximise the sum of logarithmic weights: $\sum_{i=1}^{N_t} log\left(w_t^i\right)$
- Subject to a risk constraint of target volatility $\sigma_{\boldsymbol{w}_t} \equiv \sqrt{\boldsymbol{w}_t' \cdot \boldsymbol{\Sigma}_t \cdot \boldsymbol{w}_t} \leq \sigma_{TGT}$

We first form the Lagrangian (we denote the Lagrange multiplier by λ):

$$L(\boldsymbol{w_t}) = \sum_{i=1}^{N_t} log\left(w_t^i\right) - \lambda \cdot \left(\sigma_{\boldsymbol{w}_t} - \sigma_{TGT}\right) \tag{25}$$

We then calculate all partial derivatives with respect to each weight w_t^i:

$$\frac{\partial L(\boldsymbol{w_t})}{\partial w_t^i} = \frac{1}{w_t^i} - \lambda \cdot \underbrace{\frac{\partial \sigma_{\boldsymbol{w}_t}}{\partial w_t^i}}_{\equiv MCR_t^i}, \ \ \forall i \tag{26}$$

Setting the above expression equal to zero leads to equation (24):

$$w_t^i \cdot MCR_t^i = \frac{1}{\lambda} = constant, \ \ \forall i \tag{27}$$

Post-optimisation, all weights are rescaled so that they sum up to 1. The fact that volatility exhibits positive homogeneity (for a scaling constant κ, $\sigma_{\kappa \cdot \boldsymbol{w}} = \kappa \cdot \sigma_{\boldsymbol{w}}$) renders the MCR scale-invariant as is easily deduced by:

$$\frac{\partial \sigma_{\kappa \cdot \boldsymbol{w_t}}}{\partial \left(\kappa w_t^i\right)} = \frac{\partial \sigma_{\boldsymbol{w}_t}}{\partial w_t^i} = MCR_t^i, \ \ \forall i \tag{28}$$

This means that the rescaled weights satisfy the risk-parity objective of equation (24), as the normalisation constant (the sum of unadjusted weights) is absorbed by the constant of equation (24). Hence, delaying the application of the "fully-invested" constraint until after the optimisation helps computationally and does not alter the end result in terms of the risk-parity objective.

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